

Random averages on $\mathbb{Z}^d$

Austin Eide

UofT Probability Seminar

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Random
averages on
 $\mathbb{Z}^d$

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& Duality

Proof sketch

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- Definition of the process
- Description of the main result
- Motivation and previous work
- Proof (sketch)
- Future directions

The averaging process on a finite graph

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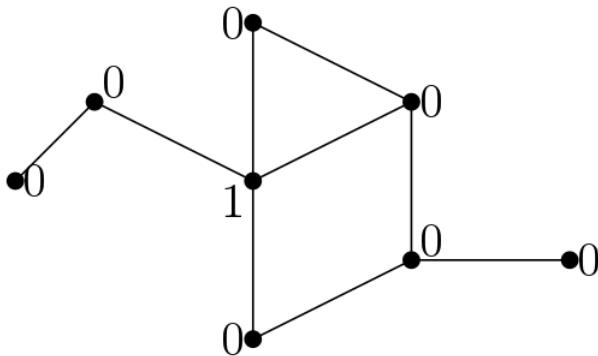
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Start with a probability distribution on vertices of G :



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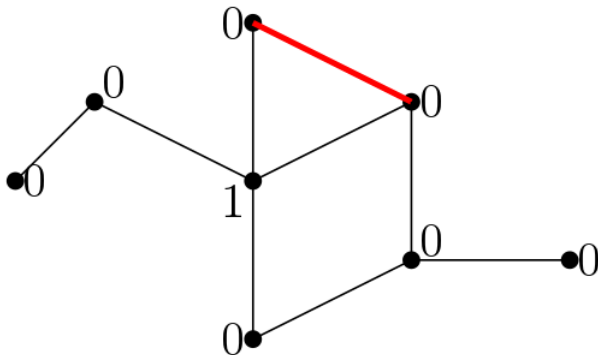
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Select edge uniformly at random, balance mass across the edge



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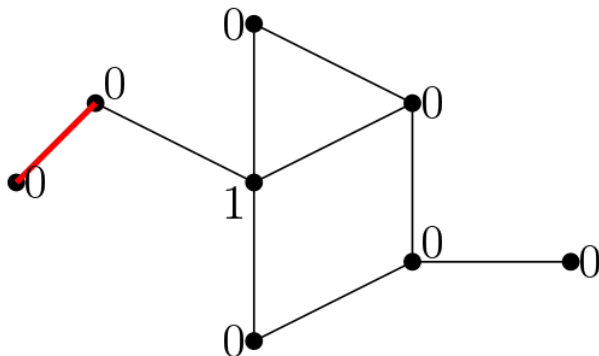
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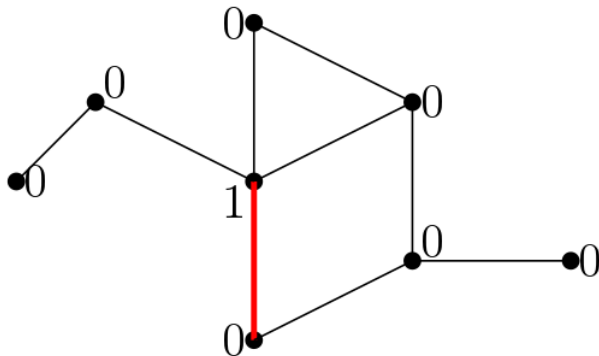
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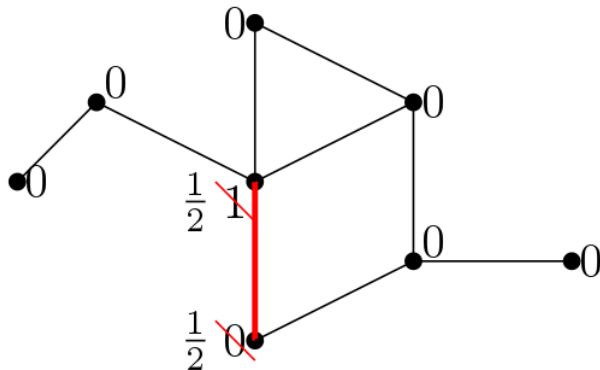
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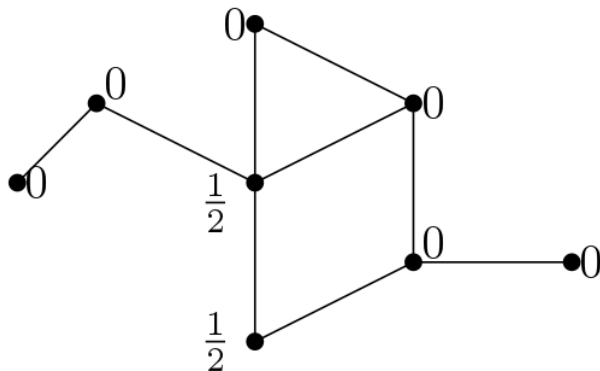
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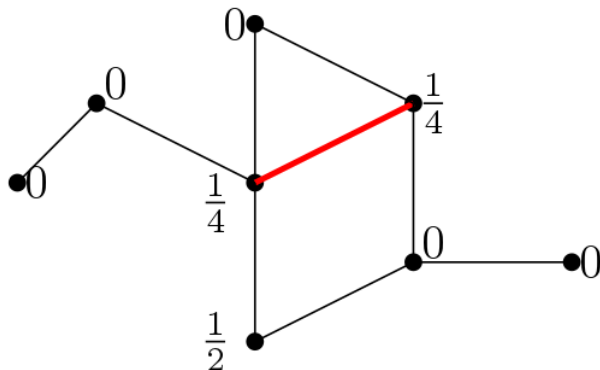
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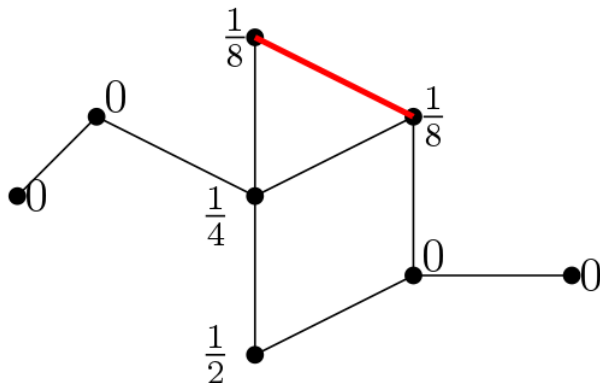
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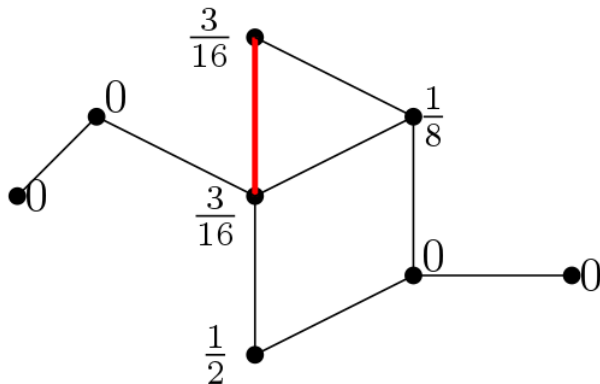
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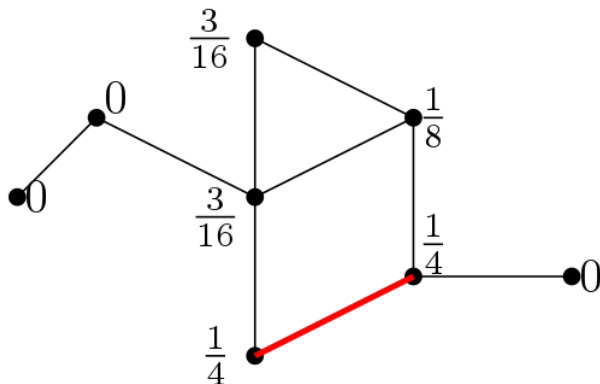
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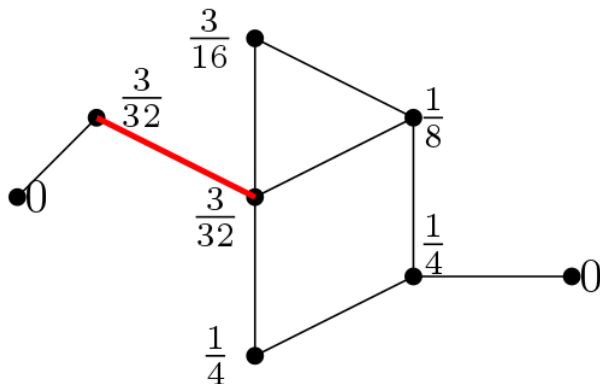
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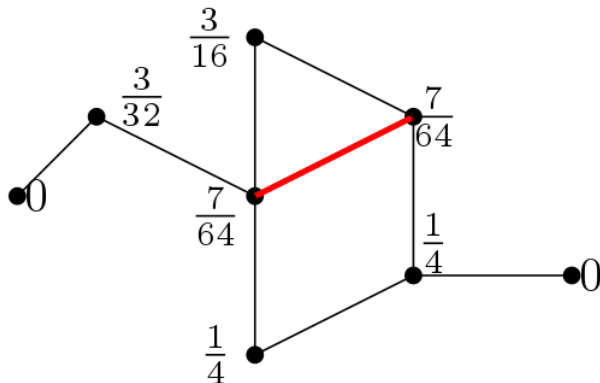
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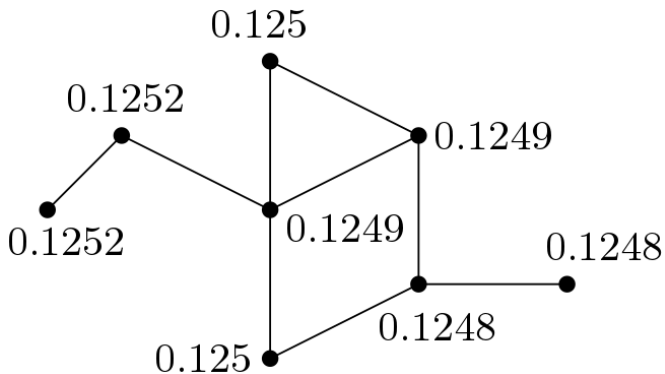
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After 150 steps:



Convergence in the finite case

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If η_t is the mass distribution after t time steps,

$$\eta_t \xrightarrow{t \rightarrow \infty} \text{Unif}(V)$$

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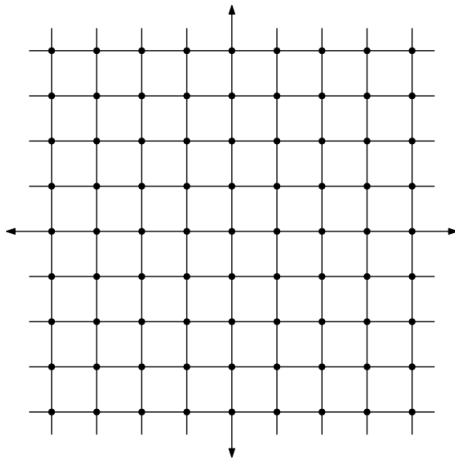
Given $\alpha > 0$, how large must $t = t(\alpha)$ be so that

$$\|\eta_t - \text{Unif}(V)\|_1 = \sum_{v \in V} \left| \eta_{v,t} - \frac{1}{|V|} \right| < \alpha?$$

(say, with high probability).

The averaging process on infinite graphs?

What if G is infinite, e.g. $\mathbb{Z}^2$?



The averaging process on infinite graphs?

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- ~~averages made according to uniformly selected edges~~ → averages made according to i.d.d. Poisson processes (“clocks”) on the edges.

The averaging process on infinite graphs?

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- ~~averages made according to uniformly selected edges~~ → averages made according to i.d.d. Poisson processes (“clocks”) on the edges. Dynamics: upon a ring on edge $\{v, w\}$, update

$$\eta_{v,t}, \eta_{w,t} = \frac{\eta_{v,t^-} + \eta_{w,t^-}}{2}.$$

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- Well-defined?

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- Well-defined?
- “Equilibrium” is 0 at each vertex, so no “mixing.”

The averaging process on infinite graphs?

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- ~~averages made according to uniformly selected edges~~ $\rightarrow$
averages made according to i.d.d. Poisson processes
("clocks") on the edges. Dynamics: upon a ring on edge
 $\{v, w\}$, update

$$\eta_{v,t}, \eta_{w,t} = \frac{\eta_{v,t^-} + \eta_{w,t^-}}{2}.$$

- Well-defined?
- "Equilibrium" is 0 at each vertex, so no "mixing." Instead, think about the profile of η_t under appropriate scaling as $t \rightarrow \infty$.

Let $(\boldsymbol{\eta}_t)$ ($t \geq 0$) be an averaging process on $\mathbb{Z}^d$ started from initial probability distribution $\boldsymbol{\eta}_0 = \boldsymbol{\eta} \in \mathbb{R}^{\mathbb{Z}^d}$.

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Theorem (d -dimensional CLT, E. '24)

For any $f \in \mathcal{C}_b(\mathbb{R}^d)$,

$$\sum_{v \in \mathbb{Z}^d} \eta_{v,t} f(v/\sqrt{t/2}) \xrightarrow{\text{Pr}} \int_{\mathbb{R}^d} f dv$$

as $t \rightarrow \infty$, where v is the d -dimensional standard Gaussian density.

Main results

 $d = 1$ Random
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Let $\Phi_t(z) = \sum_{v \leq z\sqrt{t/2}} \eta_{v,t}$ for $z \in \mathbb{R}$. (Re-scaled CDF of η_t .)

Main results

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Let $\Phi_t(z) = \sum_{v \leq z \sqrt{t/2}} \eta_{v,t}$ for $z \in \mathbb{R}$. (Re-scaled CDF of η_t .)

Theorem (1-dimensional CLT)

$$\Phi_t(z) \xrightarrow{\text{Pr}} \Phi(z)$$

as $t \rightarrow \infty$, where $\Phi(z)$ is the CDF of a standard normal random variable.

Everything is a corollary of

Theorem (ℓ^1 concentration, E. '24)

$$\|\eta_t - \mathbb{E}[\eta_t]\|_1 \xrightarrow{\text{Pr}} 0$$

as $t \rightarrow \infty$.

Everything is a corollary of

Theorem (ℓ^1 concentration, E. '24)

$$\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_1 \xrightarrow{\text{Pr}} 0$$

as $t \rightarrow \infty$. (Equivalently $\mathbb{E}[\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_1] \rightarrow 0$.)

Everything is a corollary of

Theorem (ℓ^1 concentration, E. '24)

$$\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_1 \xrightarrow{\text{Pr}} 0$$

as $t \rightarrow \infty$. (Equivalently $\mathbb{E}[\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_1] \rightarrow 0$.)

$\mathbb{E}[\boldsymbol{\eta}_t]$ is the time- t distribution of a SRW on $\mathbb{Z}^d$, hence the CLT!

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- Aldous & Lanoue (2012): *duality* with random walk, basic convergence results

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- Chatterjee, Diaconis, Sly, Zhang (2022): mixing on K_n

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- Chatterjee, Diaconis, Sly, Zhang (2022): mixing on K_n
- Quattropani, Sau, Caputo (2023): mixing on Q_n , $K_{m,n}$ (using techniques of CDSZ)
- Movassagh, Szegedy, Wang (2022): K_n exhibits fastest mixing among all finite connected graphs, results on mixing for a variety of finite graphs (star, cycle, path, etc)

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Early concentration

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Theorem (Nagahata & Yoshida '08)

On $\mathbb{Z}^d$ for $d \geq 3$, (η_t) has concentration and CLT.

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Theorem (Nagahata & Yoshida '08)

On $\mathbb{Z}^d$ for $d \geq 3$, (η_t) has concentration and CLT.

Theorem (Sau '24, Quattropani & Sau '23)

On the d -dimensional torus $(\mathbb{Z}/n\mathbb{Z})^d$, (η_t) :

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Theorem (Nagahata & Yoshida '08)

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On the d -dimensional torus $(\mathbb{Z}/n\mathbb{Z})^d$, (η_t) :

- ① mixes in time $\Theta(n^2)$

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Theorem (Nagahata & Yoshida '08)

On $\mathbb{Z}^d$ for $d \geq 3$, (η_t) has concentration and CLT.

Theorem (Sau '24, Quattropani & Sau '23)

On the d -dimensional torus $(\mathbb{Z}/n\mathbb{Z})^d$, (η_t) :

- 1 mixes in time $\Theta(n^2)$
- 2 concentrates to within $o(1)$ of $E[\eta_t]$ in time $O(n^{\frac{2d}{d+2}+\epsilon})$

One slide on existence

∞ Poisson clocks $\rightarrow$ accumulation $\rightarrow$ rings not well-ordered!

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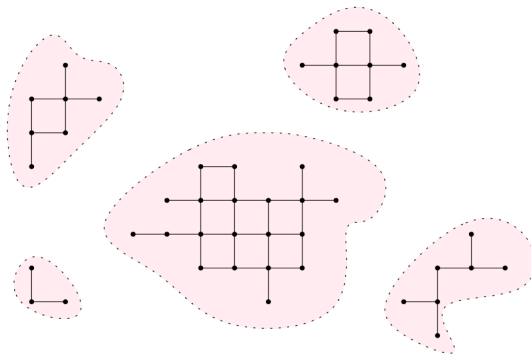
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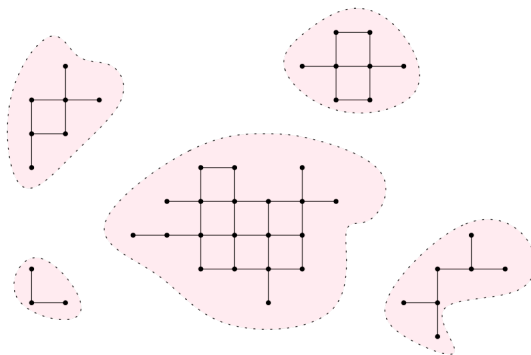
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One slide on existence

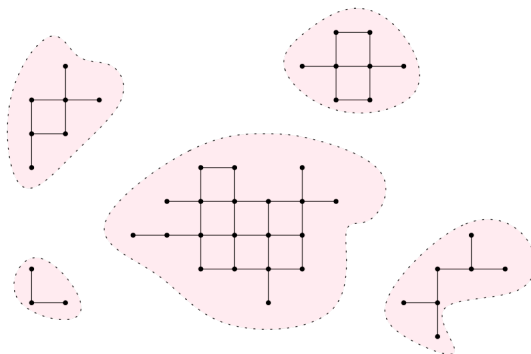
∞ Poisson clocks $\rightarrow$ accumulation $\rightarrow$ rings not well-ordered!



$$\Pr(e \text{ rings in } [0, t]) = 1 - e^{-\frac{t}{2d}}$$

One slide on existence

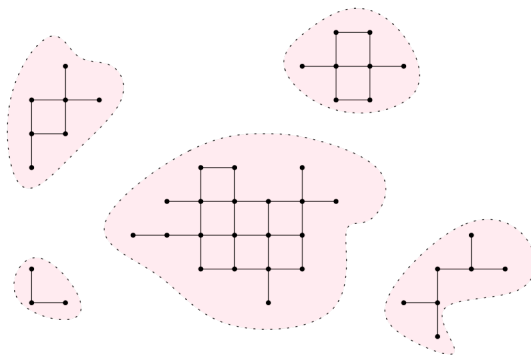
∞ Poisson clocks $\rightarrow$ accumulation $\rightarrow$ rings not well-ordered!



$$\Pr(e \text{ rings in } [0, t]) = 1 - e^{-\frac{t}{2d}} \rightarrow \text{subcrit. percolation for } t \text{ small}$$

One slide on existence

∞ Poisson clocks $\rightarrow$ accumulation $\rightarrow$ rings not well-ordered!



$\Pr(e \text{ rings in } [0, t]) = 1 - e^{-\frac{t}{2d}} \rightarrow$ subcrit. percolation for t small $\rightarrow$ finite components a.s.

Theorem

Let (W_t) ($t \geq 0$) be a CTSRW with jump rate $\frac{1}{2}$ on $\mathbb{Z}^d$.

Theorem

Let (W_t) ($t \geq 0$) be a CTSRW with jump rate $\frac{1}{2}$ on $\mathbb{Z}^d$. Then

$$\mathbb{E}[\eta_{v,t}] = \Pr(W_t = v \mid W_0 \sim \eta).$$

Theorem

Let (W_t) ($t \geq 0$) be a CTSRW with jump rate $\frac{1}{2}$ on $\mathbb{Z}^d$. Then

$$\mathbb{E}[\eta_{v,t}] = \Pr(W_t = v \mid W_0 \sim \eta).$$

Further, $\exists$ a coupling $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$ of two copies of (W_t) s.t.

$$\mathbb{E}[\eta_{v,t} \eta_{w,t}] = \Pr(\widehat{W}_t^{(1)} = v, \widehat{W}_t^{(2)} = w \mid \widehat{W}_0^{(1)}, \widehat{W}_0^{(2)} \sim \eta).$$

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Imagine a discretized version of the process in which the (continuous) mass is replaced by (discrete) tokens. Pick a special token and paint it **blue**.

Discretization

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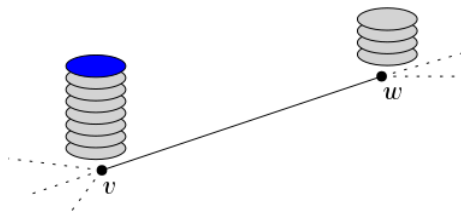
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Suppose the blue token is at v and an average across $\{v, w\}$ is to be performed



Discretization

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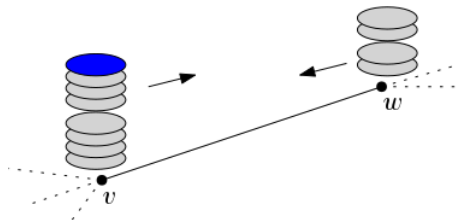
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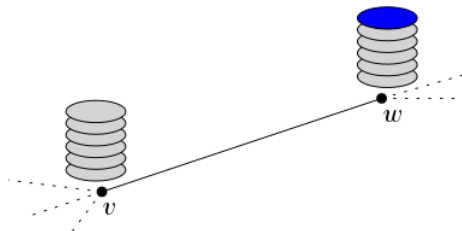
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Average can be achieved by each vertex splitting its pile and choosing a random half to send to the other vertex



The result:



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When at v , blue token jumps to neighbor w at rate $\frac{1}{4d}$ —this is the random walk (W_t) .

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When at v , **blue** token jumps to neighbor w at rate $\frac{1}{4d}$ —this is the random walk (W_t) .

If N tokens total,

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When at v , blue token jumps to neighbor w at rate $\frac{1}{4d}$ —this is the random walk (W_t) .

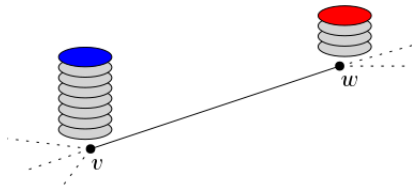
If N tokens total,

$$\begin{aligned} \mathbb{E}[\text{mass at } v \text{ at time } t] &\approx \mathbb{E} \left[\frac{1}{N} \sum_{\text{tokens}} \mathbb{1}\{\text{token at } v \text{ at time } t\} \right] \\ &= \Pr(\text{token at } v \text{ at time } t) \\ &= \Pr(W_t = v) \end{aligned}$$

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Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

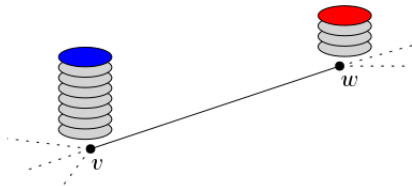
The blue and red tokens perform coupled random walks



Duality for second moments

Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

The **blue** and **red** tokens perform coupled random walks

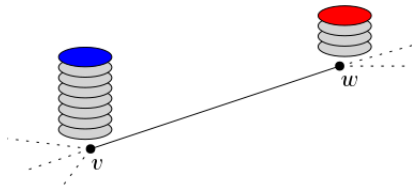


If $\{v, w\}$ rings, four possibilities:

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Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

The **blue** and **red** tokens perform coupled random walks



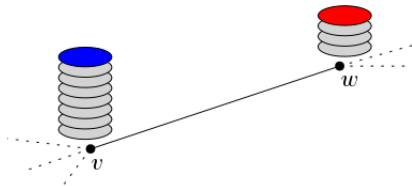
If $\{v, w\}$ rings, four possibilities:

① **blue** $\rightarrow w$, **red** stays (rate $\frac{1}{2d} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8d}$)

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Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

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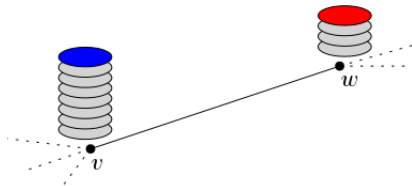
If $\{v, w\}$ rings, four possibilities:

- ① **blue** $\rightarrow w$, **red** stays (rate $\frac{1}{2d} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8d}$)
- ② **red** $\rightarrow v$, **blue** stays (rate $\frac{1}{8d}$)

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Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

The **blue** and **red** tokens perform coupled random walks



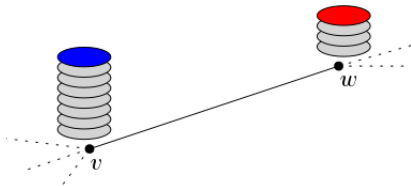
If $\{v, w\}$ rings, four possibilities:

- ① **blue** $\rightarrow w$, **red** stays (rate $\frac{1}{2d} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8d}$)
- ② **red** $\rightarrow v$, **blue** stays (rate $\frac{1}{8d}$)
- ③ **blue** $\rightarrow w$, **red** $\rightarrow v$ (rate $\frac{1}{8d}$)

Duality for second moments

Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

The **blue** and **red** tokens perform coupled random walks



If $\{v, w\}$ rings, four possibilities:

- 1 **blue** $\rightarrow w$, **red** stays (rate $\frac{1}{2d} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8d}$)
- 2 **red** $\rightarrow v$, **blue** stays (rate $\frac{1}{8d}$)
- 3 **blue** $\rightarrow w$, **red** $\rightarrow v$ (rate $\frac{1}{8d}$)
- 4 **blue** and **red** stay (rate $\frac{1}{8d}$).

Duality for second moments

The difference walk ($\hat{D}_t$)

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Rest of dynamics worked out similarly.

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Rest of dynamics worked out similarly. Useful to consider $\hat{D}_t = \widehat{W}_t^{(1)} - \widehat{W}_t^{(2)}$ (also a walk on $\mathbb{Z}^d$!):

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The difference walk ($\hat{D}_t$)

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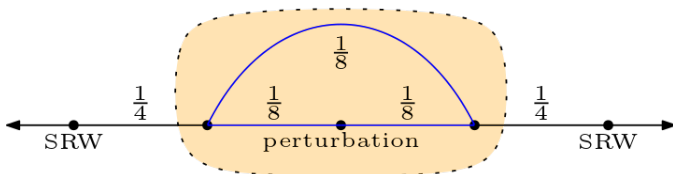
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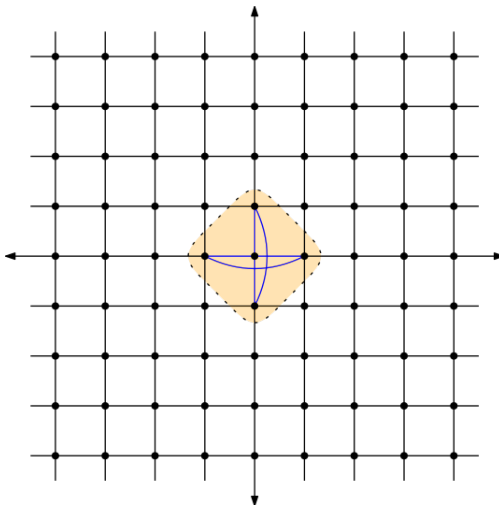
Rest of dynamics worked out similarly. Useful to consider $\hat{D}_t = \widehat{W}_t^{(1)} - \widehat{W}_t^{(2)}$ (also a walk on $\mathbb{Z}^d$):



Duality for second moments

The difference walk ($\hat{D}_t$)

In two dimensions:



Duality for second moments

Dynamics for $(\widehat{W}_t^{(1)}, \widehat{W}_t^{(2)})$

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In general, $(\widehat{D}_t)$ is a CTSRW with jump rate 1 and a perturbation at vertices distance ≤ 1 from the origin.

Identity for ℓ^2 norm

Goal:

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Identity for ℓ^2 norm

Goal: $E [\|\eta_t - E[\eta_t]\|_1] \rightarrow 0.$

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Identity for ℓ^2 norm

Goal: $E [\|\eta_t - E[\eta_t]\|_1] \rightarrow 0$. Identity:

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Identity for ℓ^2 norm

Goal: $\mathbb{E} [\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_1] \rightarrow 0$. Identity:

$$\begin{aligned}\mathbb{E} \left[\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_2^2 \right] &= \mathbb{E} \left[\sum_{v \in \mathbb{Z}^d} (\eta_{v,t} - \mathbb{E}[\eta_{v,t}])^2 \right] \\ &= \sum_{v \in \mathbb{Z}^d} (\mathbb{E}[\eta_{v,t}^2] - \mathbb{E}[\eta_{v,t}]^2) \\ &= \sum_{v \in \mathbb{Z}^d} (\Pr(\widehat{W}_t^{(1)} = \widehat{W}_t^{(2)} = v) - \Pr(W_t = v)^2) \\ &= \Pr(\widehat{D}_t = \mathbf{0}) - \Pr(D_t = \mathbf{0})\end{aligned}$$

Identity for ℓ^2 norm

Goal: $\mathbb{E} [\|\eta_t - \mathbb{E}[\eta_t]\|_1] \rightarrow 0$. Identity:

$$\begin{aligned}
 \mathbb{E} [\|\eta_t - \mathbb{E}[\eta_t]\|_2^2] &= \mathbb{E} \left[\sum_{v \in \mathbb{Z}^d} (\eta_{v,t} - \mathbb{E}[\eta_{v,t}])^2 \right] \\
 &= \sum_{v \in \mathbb{Z}^d} (\mathbb{E}[\eta_{v,t}^2] - \mathbb{E}[\eta_{v,t}]^2) \\
 &= \sum_{v \in \mathbb{Z}^d} (\Pr(\widehat{W}_t^{(1)} = \widehat{W}_t^{(2)} = v) - \Pr(W_t = v)^2) \\
 &= \Pr(\widehat{D}_t = \mathbf{0}) - \Pr(D_t = \mathbf{0})
 \end{aligned}$$

Note: $(D_t) = (W_t^{(1)} - W_t^{(2)})$ is a CTSRW on $\mathbb{Z}^d$ (no perturbation).

Cauchy-Schwarz

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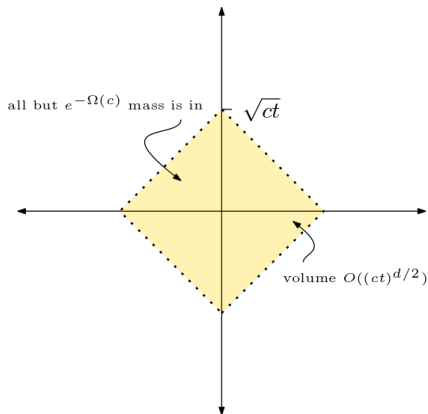
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Cauchy-Schwarz

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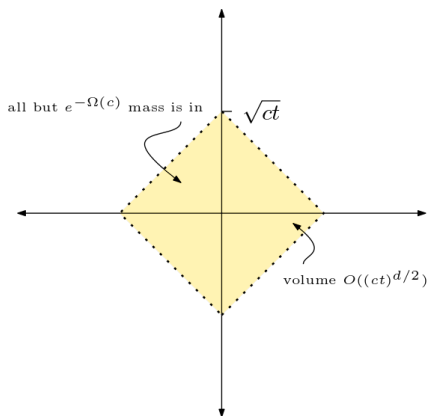
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$$\mathbb{E} \left[\|\eta_t - \mathbb{E}[\eta_t]\|_1^2 \right] \stackrel{\text{C-S}}{\leq} O \left(t^{d/2} \mathbb{E} \left[\|\eta_t - \mathbb{E}[\eta_t]\|_2^2 \right] \right)$$

Cauchy-Schwarz

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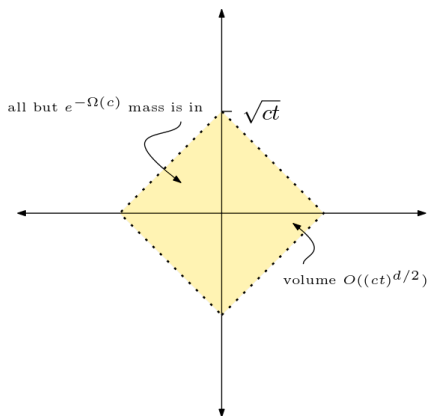
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$$\mathbb{E} \left[\|\eta_t - \mathbb{E}[\eta_t]\|_1^2 \right] \stackrel{\text{C-S}}{=} O \left(t^{d/2} \mathbb{E} \left[\|\eta_t - \mathbb{E}[\eta_t]\|_2^2 \right] \right) = O \left(t^{d/2} \left(\Pr(\hat{D}_t = \mathbf{0}) - \Pr(D_t = \mathbf{0}) \right) \right)$$

Reduction to local limit theorem

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Need to show $t^{d/2} \left(\Pr(\widehat{D}_t = \mathbf{0}) - \Pr(D_t = \mathbf{0}) \right) \rightarrow 0$.

Reduction to local limit theorem

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Need to show $t^{d/2} \left(\Pr(\widehat{D}_t = \mathbf{0}) - \Pr(D_t = \mathbf{0}) \right) \rightarrow 0$.

Crucial: both $\Pr(\widehat{D}_t = \mathbf{0})$ and $\Pr(D_t = \mathbf{0})$ are $\Theta(t^{-d/2})$!

So need exact expressions for leading terms ((D_t) is known from usual LLT).

Theorem

As $t \rightarrow \infty$,

$$\Pr(\hat{D}_t = \mathbf{0}), \Pr(D_t = \mathbf{0}) = (1 + o(1)) \frac{1}{(2\pi t)^{d/2}}$$

Theorem

As $t \rightarrow \infty$,

$$\Pr(\widehat{D}_t = \mathbf{0}), \Pr(D_t = \mathbf{0}) = (1 + o(1)) \frac{1}{(2\pi t)^{d/2}}$$

To prove,

- 1 Switch to discrete time
- 2 Find expression relating probability generating functions
- 3 Singularity analysis
- 4 Switch back to continuous time

Local limit theorem

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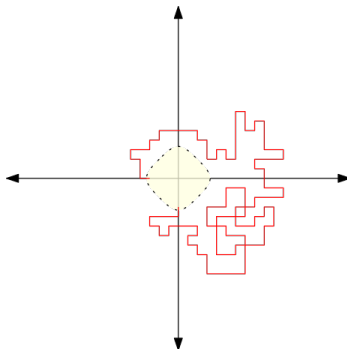
Proof sketch

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$$G(z) = \sum_{n=0}^{\infty} \Pr(D_n = \mathbf{0}) z^n \text{ and } \hat{G}(z) = \sum_{n=0}^{\infty} \Pr(\hat{D}_n = \mathbf{0}) z^n$$

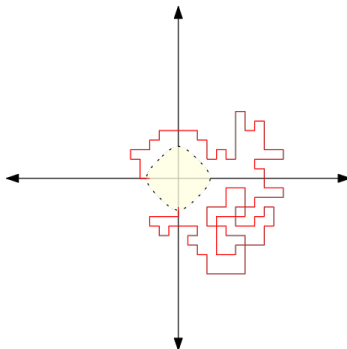
Local limit theorem

$$G(z) = \sum_{n=0}^{\infty} \Pr(D_n = \mathbf{0}) z^n \text{ and } \hat{G}(z) = \sum_{n=0}^{\infty} \Pr(\hat{D}_n = \mathbf{0}) z^n$$



Local limit theorem

$$G(z) = \sum_{n=0}^{\infty} \Pr(D_n = \mathbf{0}) z^n \text{ and } \hat{G}(z) = \sum_{n=0}^{\infty} \Pr(\hat{D}_n = \mathbf{0}) z^n$$



Can write $G(z) = A(S(z))$ and $\hat{G}(z) = B(S(z))$, where $S(z)$ corresponds to excursions which begin and end on the ℓ^1 unit sphere and don't visit $\mathbf{0}$.

Remaining questions

General stochastic flows

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Whole class of processes characterized by

$$\eta_t = A_t \eta_{t-}$$

where (A_t) is a matrix process in $\mathbb{R}^{\mathbb{Z}^d \times \mathbb{Z}^d}$.

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Whole class of processes characterized by

$$\eta_t = A_t \eta_{t-}$$

where (A_t) is a matrix process in $\mathbb{R}^{\mathbb{Z}^d \times \mathbb{Z}^d}$. Can we do the same trick?

Potlach process

a.k.a. meteor process

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Each vertex is hit by a meteor at rate 1, all mass distributes evenly among neighbors.

Potlach process

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Each vertex is hit by a meteor at rate 1, all mass distributes evenly among neighbors.

$$\Pr(\hat{D}_t = \mathbf{0}) = 2(1 + o(1))\Pr(D_t = \mathbf{0})$$

Potlach process

a.k.a. meteor process

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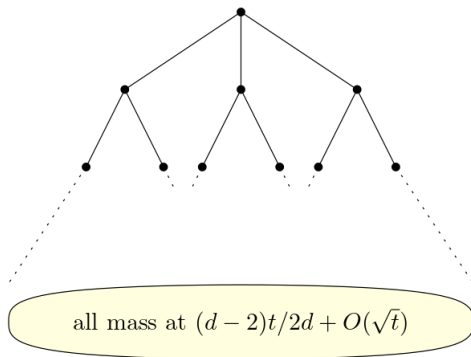


Each vertex is hit by a meteor at rate 1, all mass distributes evenly among neighbors.

$$\Pr(\hat{D}_t = \mathbf{0}) = 2(1 + o(1))\Pr(D_t = \mathbf{0})$$

$$\implies \mathbb{E} \left[\|\boldsymbol{\eta}_t - \mathbb{E}[\boldsymbol{\eta}_t]\|_2^2 \right] = \Theta(t^{-d/2})$$

Random averages on regular d -ary tree?



How is mass distributed among the vertices at depth $\approx \frac{(d-2)t}{2d}$?

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Questions?